The Banach-Mazur distance between the cube and the cross-polytope in dimensions three and four

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Introduction

We introduce some definitions:

- A set $K \subseteq \mathbb{R}^n$ is called a *convex body* if it is compact, convex, and has a non-empty interior.
- The *Banach-Mazur distance* between two convex bodies $K, L \subseteq \mathbb{R}^n$, both centrally symmetric around 0 is

$$d_{BM}(K, L) := \inf\{r > 0 : K \subseteq TL \subseteq rK\}$$

  where the infimum is taken over all invertible linear operators $T : \mathbb{R}^n \to \mathbb{R}^n$.
- C_n is the *n-dimensional hypercube*, the convex body with 2^n vertices $\epsilon \in \{-1, 1\}^n$. Equivalently, $C_n = \{x \in \mathbb{R}^n : \|x\|_\infty \leq 1\}$ where $x = (x_1, ..., x_n)$ and $\|x\|_\infty = \max\{x_1, ..., x_n\}$.
- C_n^* is the *n-dimensional cross-polytope*, the convex body with $2n$ vertices $(0, ..., 0, \pm 1, 0, ..., 0) \in \mathbb{R}^n$. Equivalently, it is the dual of C_n.

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Computing $d_{BM}(C_n, C_n^*)$

The aim of this talk is to compute $d_{BM}(C_n, C_n^*)$ for $n = 3, 4$.

Note that

$$\frac{1}{d_{BM}(K,L)} = \sup\{r > 0 : rK \subseteq TL \subseteq K \text{ for some } T \in M_{n\times n}(\mathbb{R})\},$$

since $K \subseteq TL \subseteq sK \iff \frac{1}{s}L \subseteq T^{-1}K \subseteq L$.

Applying this formula to C_n, C_n^*, we obtain

$$\frac{1}{d} = \sup\{r \in \mathbb{R}^+ : rC_n \subseteq TC_n^* \subseteq C_n \text{ for some operator } T \in M_{n\times n}(\mathbb{R})\}.$$

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For the $rC_n \subseteq TC_n^*$ inclusion, we can restate it as

$$rC_n \subseteq TC_n^* \iff \left\|T^{-1}w\right\|_1 \leq \frac{1}{r} \; \forall w \in \texttt{Vertex}(C_n).$$

Note that $\texttt{Vertex}(C_n) = \{-1, 1\}^n$.

Finding the Banach-Mazur distance is equivalent to finding the minimum of the function $f(T) := \max\{\left\|T^{-1}w\right\|_1 : w \in \texttt{Vertex}(C_n)\}$ over the space of all invertible operators with entries $|T_{ij}| \leq 1$. This can be stated as

$$d_{BM}(C_n, C_n^*) = \inf\{f(T) \; : |T_{ij}| \leq 1 \; \forall i, j; \; \det T \neq 0\}.$$

Ignoring the invertibility requirement, f is essentially a function $[-1, 1]^{n^2} \to \mathbb{R}^+$. We can experimentally find the minimum using computational methods, at least in low dimensions.

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Computational results

- $n = 3$ $T = \begin{bmatrix} \frac{1}{3} & -1 & -1 \\ -1 & \frac{1}{3} & -1 \\ -1 & -1 & \frac{1}{3} \end{bmatrix}$ $f(T) = \frac{9}{5}$ Proven optimal: ✓
- $n = 4$

$$T = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & -1 & -1 \\ 0 & -1 & 1 & -1 \\ 0 & 1 & 1 & -1 \end{bmatrix}, \begin{bmatrix} -1 & 1 & 0 & -1 \\ -1 & -1 & -1 & 0 \\ 1 & 0 & -1 & -1 \\ 0 & 1 & -1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \end{bmatrix}$$

f(T) = 2 Proven optimal: ✓

- $n = 5$

$$T = \begin{bmatrix} 0.792559 & 1 & 0.0387439 & 1 & 0.704555 \\ 1 & 0.792092 & 0.999411 & 0.855944 & 1 \\ 1 & 0.0773263 & 1 & 1 & 0.888962 \\ 0.925403 & 1 & 1 & 0.115724 & 0.822648 \\ 1 & 0.79255 & 0.999989 & 0.856439 & 1 \end{bmatrix}$$

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Summary of computational results

n	2	3	4	5	6	7	8
$d_{BM}(C_n, C_n^*)$	1	1.8	2	2.32871	2.4488	2.6	2.5

- There is no "obvious" pattern for the optimal operator in $\mathbb{R}^n$ nor the value $d_{BM}(C_n, C_n^*)$. Although, dimensions $n = 7, 8$ seem to have optimal operators with entries from $\{-1, 0, 1\}$.
- Because the number of variables is n^2, minimising f is only feasible for small n, at least without further insight.
- For the code, the exact operators achieving the minimum, and the calculated values of f up to dimension 8, see "On the Banach-Mazur Distance between the Cube and the Crosspolytope" by Fei Xue[1].

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The BM distance in $\mathbb{R}^3$

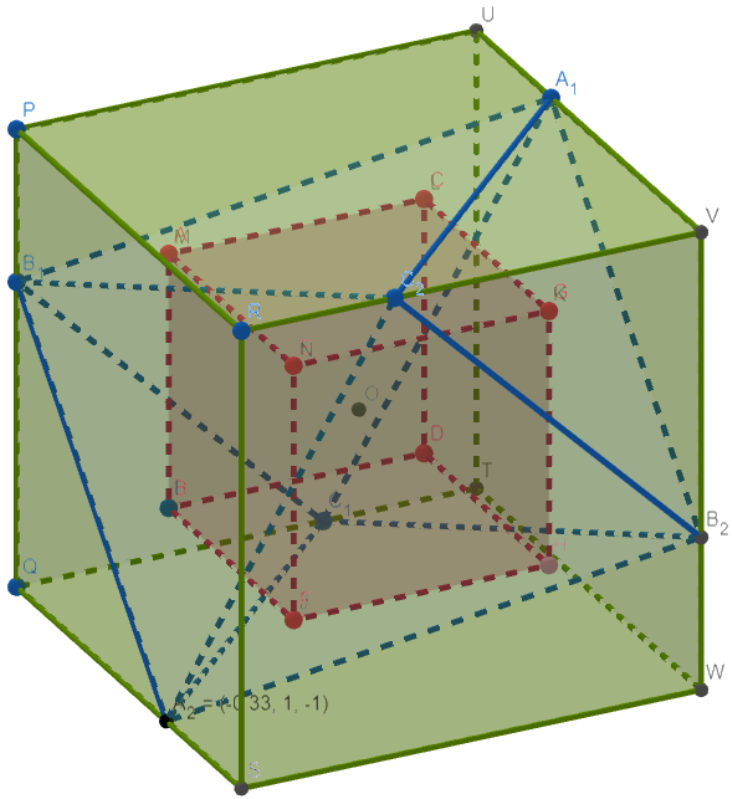
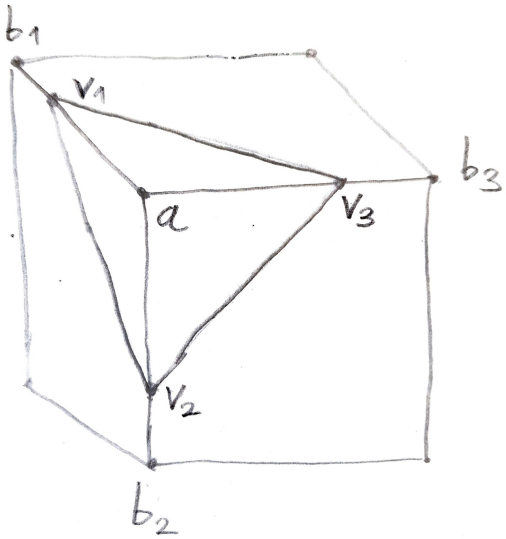
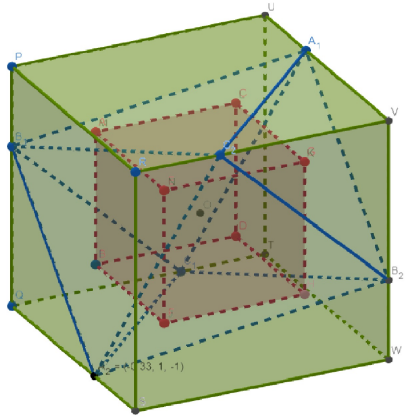


Figure: Geometric view of $\frac{5}{9}C \subseteq TC^* \subseteq C$ for $T = \begin{bmatrix} \frac{1}{3} & -1 & -1 \\ -1 & \frac{1}{3} & -1 \\ -1 & -1 & \frac{1}{3} \end{bmatrix}$.

Green cube is C, red cube is $\frac{5}{9}C$ and blue octahedron is TC^*.

A "nice" class of operators

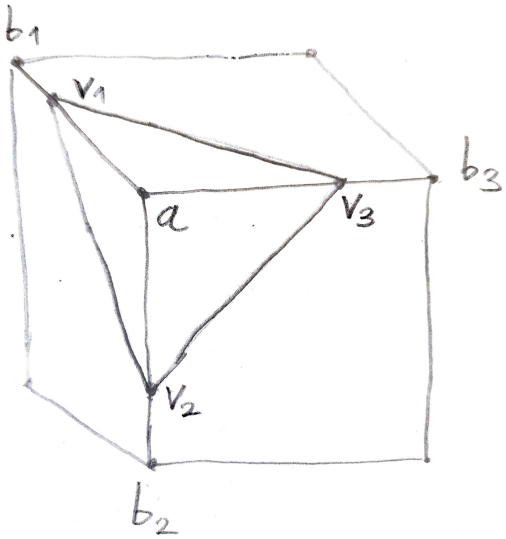


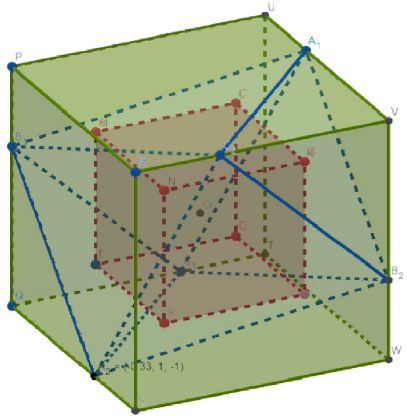


A *nice* octahedron is constructed by choosing any vertex a of the unit cube and defining $v_i := \frac{1}{3}a + \frac{2}{3}b_i$ for $i = 1, 2, 3$ where b_i are vertices adjacent to a.

This nice octahedron can be written as $\mathrm{Conv}\{v_1, v_2, v_3, -v_1, -v_2, -v_3\} = TC^*$ where $T = \begin{bmatrix} v1 & v2 & v3 \end{bmatrix}$. We call such operators T *nice* as well. There are exactly 192 nice operators. All nice operators satisfy $\frac{5}{9}C \subseteq TC^* \subseteq C$.

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Characterizing optimal operators

Our goal is to prove that

$$\frac{5}{9}C \subseteq TC^* \subseteq C \implies TC^* \text{is nice.} \tag{1}$$

This is sufficient for $d_{BM}(C, C^*) = \frac{9}{5}$ because for nice octahedrons TC^*, we easily check that $rC \not\subseteq TC^*$ if $r > \frac{5}{9}$.

From here on, we will be assuming that T satisfies $\frac{5}{9}C \subseteq TC^* \subseteq C$.

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Looking at the problem in $\mathbb{R}^2$

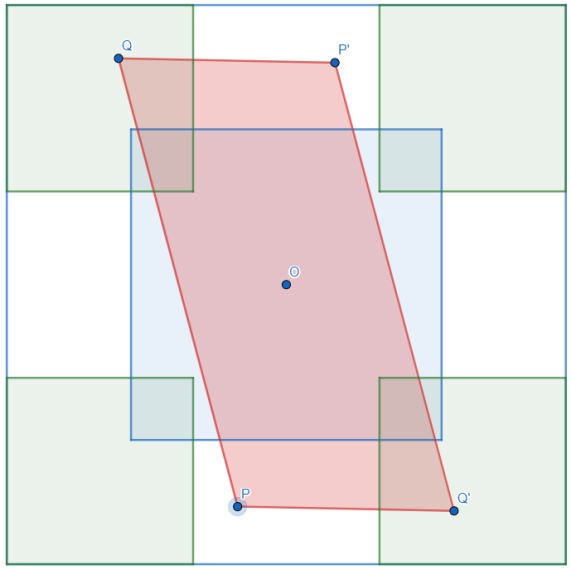


Figure: $PQP'Q'$ (red) is a parallelogram $\mathcal{R}$ symmetric around 0 contained in C_2. The blue inner square is $\frac{5}{9}C_2$. The green area are points (x, y) such that $1 \geq |x| \geq \frac{1}{3}$ and $1 \geq |y| \geq \frac{1}{3}$. Note that green area is made up of 4 green squares.

Lemma 1

If $\frac{5}{9}C_2 \subseteq \mathcal{R} \subseteq C_2$ then each of the 4 vertices of $\mathcal{R}$ lies in its own green square.

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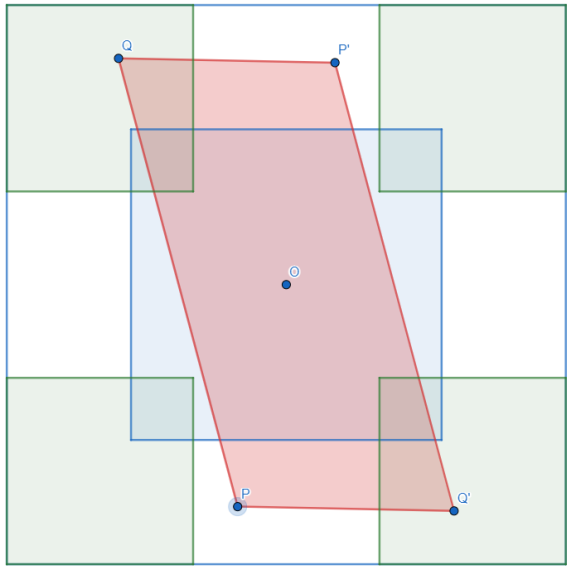


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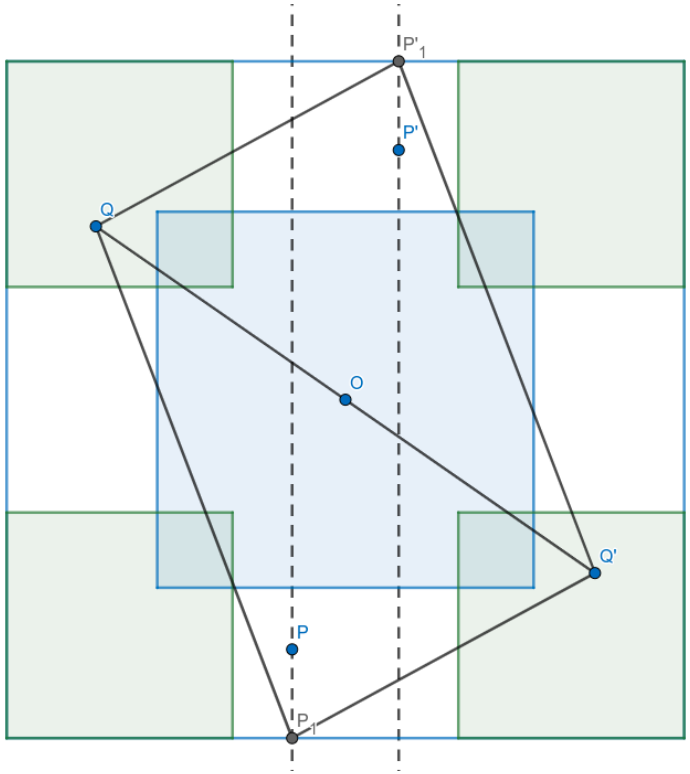
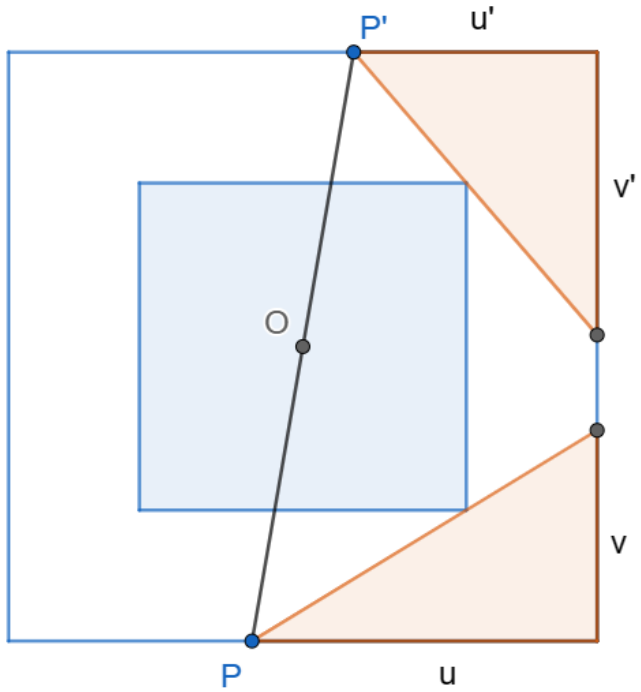


Figure: Let us assume that $\frac{5}{9}C_2 \subseteq \mathcal{R} \subseteq C_2$ and that a vertex P of $\mathcal{R}$ is not in the green area. If P_1 is the projection of P to the corresponding side of C_2 then the segment QQ' cuts the line PP_1. Therefore the the parallelogram $P_1P_1'QQ'$ contains $\mathcal{R}$. Thus we can assume that P is on the border of C_2.

Proof of Lemma 1



We assumed that $u, u' \in (\frac{2}{3}, \frac{4}{3})$. We have $u + u' = 2, v = \frac{4u}{9u-4}, v' = \frac{4v'}{9v'-4}$ thus

$$2 = v + v' = \frac{8}{9} + \frac{9}{16} \frac{10}{(9u-4)(14-9u)} < 2$$

where the inequality holds because $(9u-4)(14-9u) > 16$ for $u \in (\frac{2}{3}, \frac{4}{3})$. We arrive at a contradiction.

Proof of Lemma 1

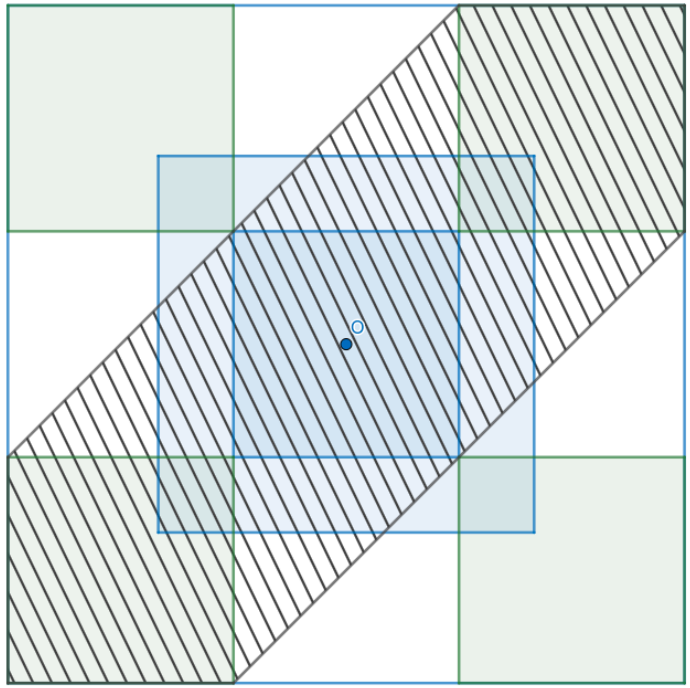


Figure: For the second part of Lemma 1, where all vertices are in some green area, we notice that if two vertices of $\mathcal{R}$ are contained in the same green square, $\mathcal{R}$ is then a subset of the black shaded figure or the analogous figure along the other diagonal of C_2. Because this figure doesn't contain $\frac{5}{9}C_2$, we conclude that if $\frac{5}{9}C_2 \subseteq \mathcal{R}$ then each vertex of $\mathcal{R}$ is in its own green square. $\square$

Back to the original problem

We start by noting that the set of points $\{(x, y, z) \in \mathbb{R}^3 : \frac{1}{3} \leq |x|, |y|, |z| \leq 1\}$ is made up of 8 disjoint closed cubes, each containing an unique vertex of C_3.

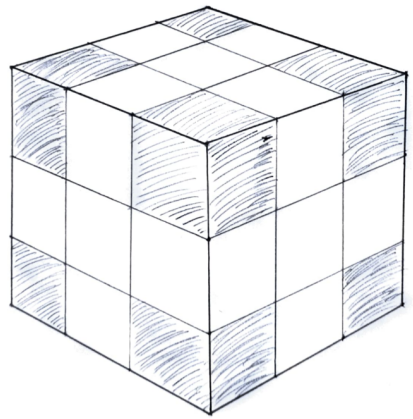


Figure: We will call these 8 cubes **corner boxes**.

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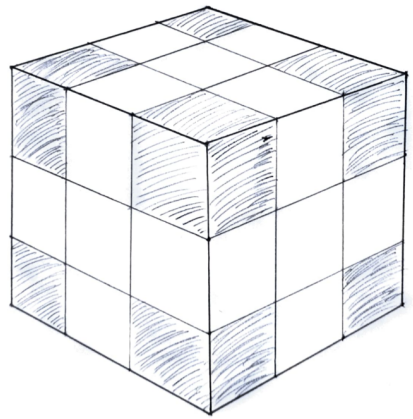


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Each vertex of TC^* is contained in a corner box. Moreover, no two vertices of TC^* are in the same corner box.

The proof of Lemma 2

Let e_1, e_2, e_3 be the standard basis in $\mathbb{R}^3$. Let $H \subset \mathbb{R}^3$ be a 2-dimensional subspace of $\mathbb{R}^3$ such that $e_i \in H$ for some $i = 1, 2, 3$.

Key observation

The orthogonal projection of C_3 to H is a rectangle, the orthogonal projection of $\frac{5}{9}C_3$ is this rectangle scaled by $\frac{5}{9}$ and the i-th coordinate is preserved by this projection.

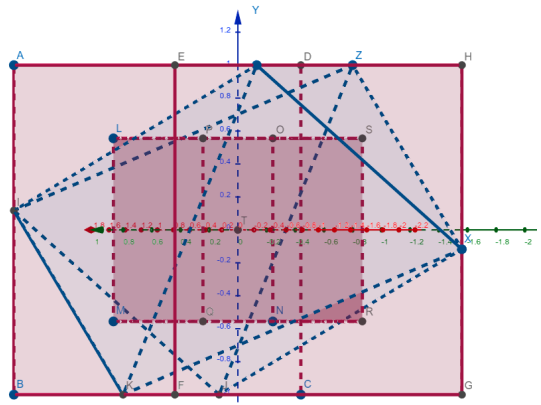


Figure: Example of such projection to a subspace H_2 containing e_2. The 2nd or the y coordinate is preserved.

The proof of Lemma 2

Choose v_1, v_2, v_3 to be different, non opposite vertices of TC^* such that $0 \leq z_1 \leq z_2 \leq z_3$ where $v_j = (x_j, y_j, z_j)$ for $j = 1, 2, 3$.
Let f_1 be the vectors perpendicular to the planes through v_1, v_2, v_3.
Finally, let H_1 be the 2-dimensional subspace containing e_3 and f_1.

We now focus at the orthogonal projection to H_1. Notice that the plane through v_1, v_2, v_3 is projected to a line. Since $0 \leq z_1 \leq z_2 \leq z_3$ and the z coordinate is preserved under projection, v_2' is between v_1' and v_3' on this line or $v_2' \in [v_1', v_3']$. Because $\pm v_1, \pm v_2, \pm v_3$ are the vertices of TC^*, we conclude that $(TC^*)' = \texttt{Conv}\{\pm v_1', \pm v_3'\}$. We also have $\frac{5}{9} C' \subseteq (TC^*)' \subseteq C'$ so by Lemma 1, we conclude $|z_1|, |z_3| \geq \frac{1}{3}$ which means $z_1, z_3 \geq \frac{1}{3}$ because they are nonnegative.

We conclude $z_j \geq \min\{z_1, z_2, z_3\} = z_1 \geq \frac{1}{3}$ for $j = 1, 2, 3$. Thus each vertex of TC^* has a z-coordinate with absolute value greater or equal than $\frac{1}{3}$. By applying the same reasoning to x and y coordinates, we conclude that each vertex of TC^* is in some corner box.

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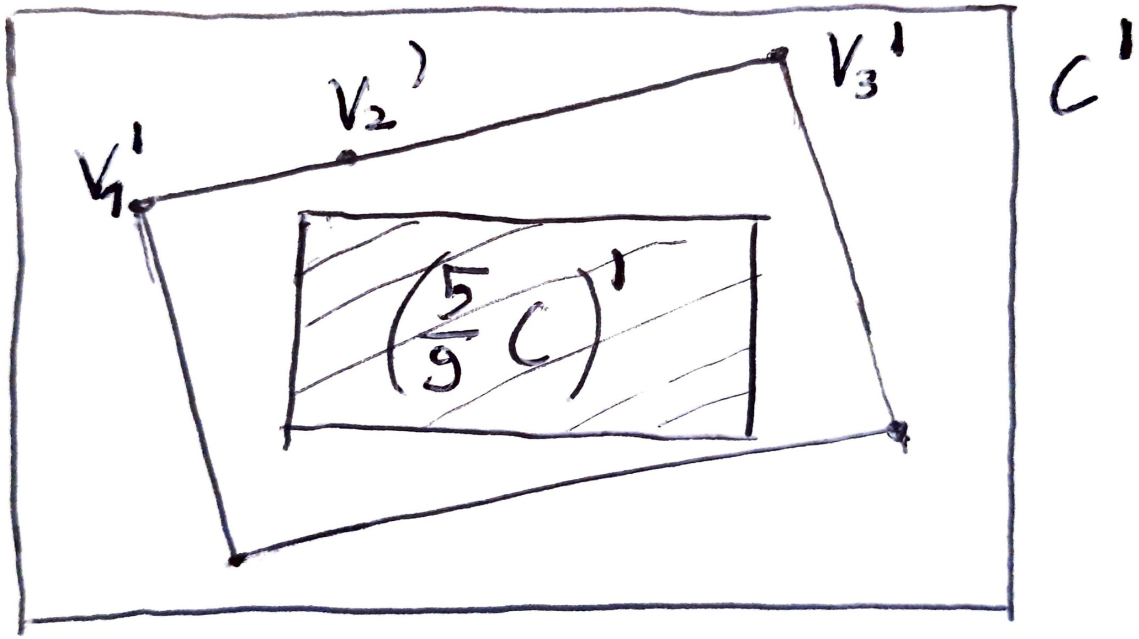


Figure: Illustration of orthogonal projection to H_1.

The proof of Lemma 2

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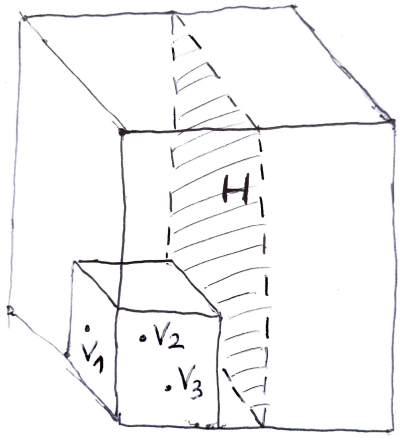


Figure: **Case 1:** Some three vertices v_1, v_2, v_3 of TC^* are in the same corner box. We choose H to be any 2-dimensional subspace containing e_{i_1}, e_{i_2} $i_1 \neq i_2$.

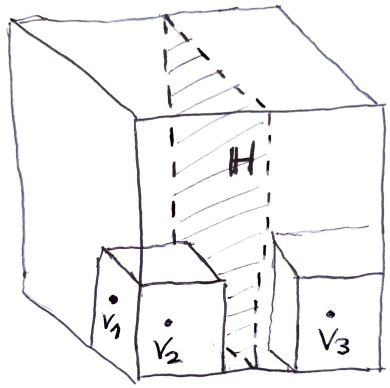


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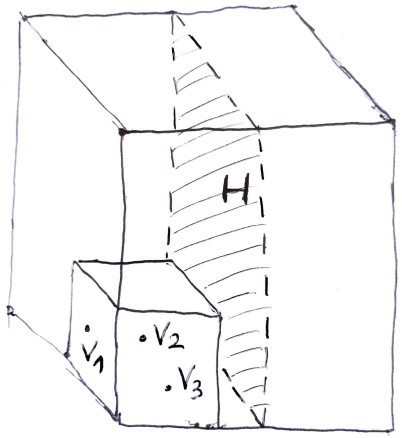


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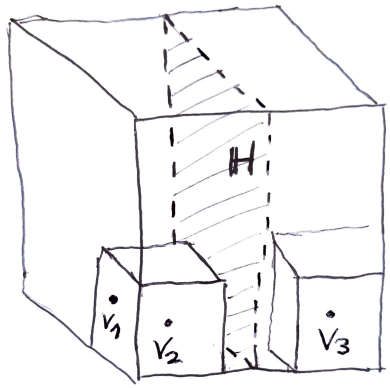


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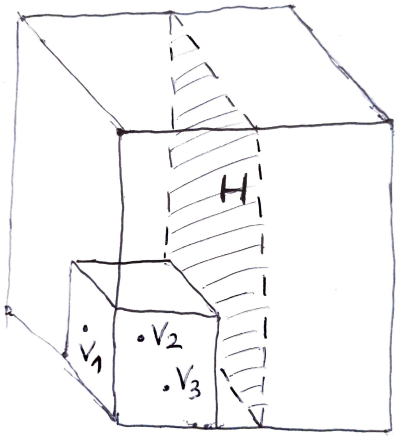


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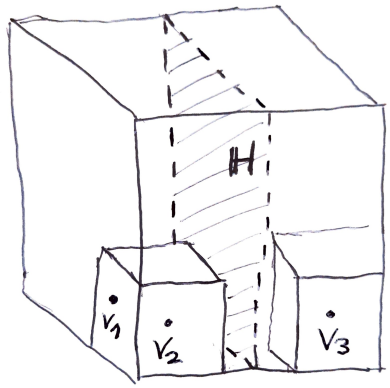


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The proof of Lemma 2

Now we consider the orthogonal projection to the chosen subspace H. We note that C_3 is projected to C_2 so we can define green areas in H and we notice that the points v_1, v_2, v_3 all get projected to the same green area.

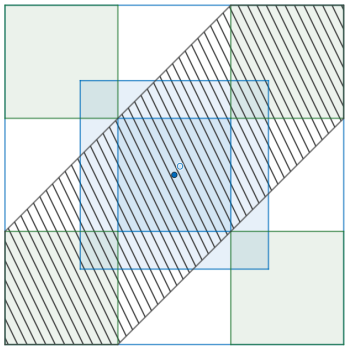


Figure: Repeating the same argument from Lemma 1, we see that $(TC^*)' = \mathtt{Conv}\{\pm v_1', \pm v_2', \pm v_3'\}$ doesn't contain $(\frac{5}{9}C_3)' = \frac{5}{9}C_2$ as it is contained in either the shaded area in the figure or in the analogous area along the other diagonal of C_2.

$(\frac{5}{9}C_3)' \not\subseteq (TC^*)' \implies \frac{5}{9}C_3 \not\subseteq TC^*$ which is a contradiction. This finishes the proof of Lemma 2. $\square$

Concluding the 3-dimensional proof

There are 8 corner boxes and 6 vertices of TC^*, therefore there exists an empty corner box.

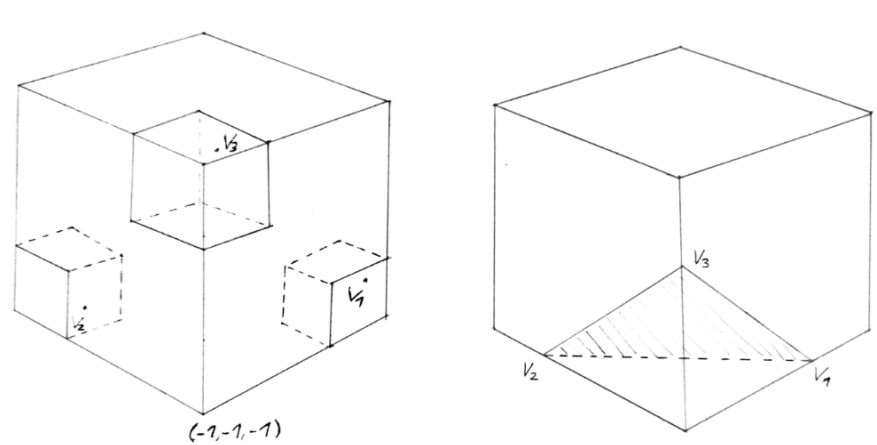


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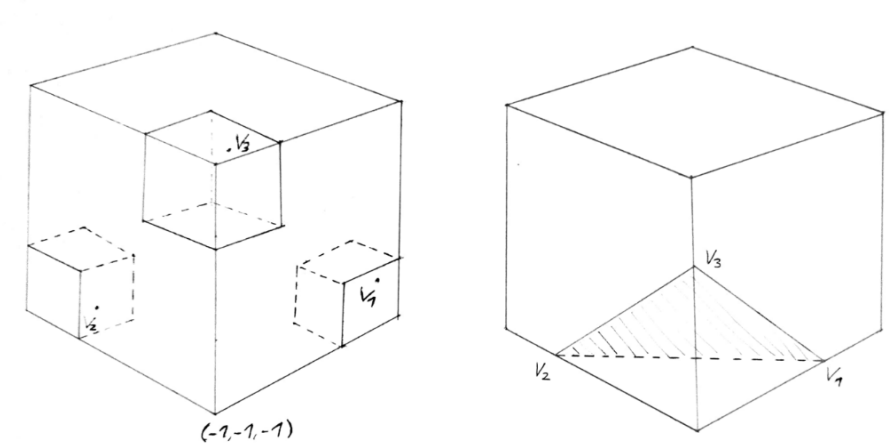


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The vertices are

$$v_1 = (x_1, y_1, z_1) \quad x_1 \geq \frac{1}{3}, y_1 \leq \frac{-1}{3}, z_1 \leq \frac{-1}{3}$$
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$$\frac{5}{9}C_3 \subseteq TC^* \implies ||\frac{1}{3}v_1 + \frac{1}{3}v_2 + \frac{1}{3}v_3||_\infty \geq \frac{5}{9}$$

WLOG we assume $|\frac{1}{3}z_1 + \frac{1}{3}z_2 + \frac{1}{3}z_3| \geq \frac{5}{9}$. Due to $\frac{1}{3}(z_1 + z_2 + z_3) \leq \frac{1}{3}(1 - \frac{1}{3} - \frac{1}{3} = \frac{1}{9} < \frac{5}{9}$, it follows that

$$\frac{-5}{9} \geq \frac{1}{3}(z_1 + z_2 + z_3) \geq \frac{1}{3}(-1 - 1 + \frac{1}{3}) = \frac{-5}{9}$$

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For small $\epsilon > 0$, $\frac{5}{9} - \frac{4}{3}\epsilon = |(\frac{1}{3} - \frac{\epsilon}{2})z_1 + (\frac{1}{3} - \frac{\epsilon}{2})z_2 + (\frac{1}{3} + \epsilon)z_3| < \frac{5}{9}$ but $||(\frac{1}{3} - \frac{\epsilon}{2})v_1 + (\frac{1}{3} - \frac{\epsilon}{2})v_2 + (\frac{1}{3} + \epsilon)v_3||_\infty \geq \frac{5}{9} \implies x$ or y achieves $|\cdot| \geq \frac{5}{9}$

For a sequence $\epsilon_k \to 0$, one of x, y achieves $|\cdot| \geq \frac{5}{9}$ for infinitely many k. WLOG let this be y. So taking the limit we obtain

$$|(\frac{1}{3} - \frac{\epsilon_k}{2})y_1 + (\frac{1}{3} - \frac{\epsilon_k}{2})y_2 + (\frac{1}{3} + \epsilon_k)y_3| \to |\frac{1}{3}y_1 + \frac{1}{3}y_2 + \frac{1}{3}y_3| \geq \frac{5}{9}.$$

Arguing similarly as we did for z, we obtain $\boxed{y_1 = -1, y_2 = \frac{1}{3}, y_3 = -1}$

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Moving on to $\mathbb{R}^4$ - pretty pictures

My first attempt at proving $d_{BM}(C_4, C_4^*) = 2$ was to look at projections to $\mathbb{R}^3$ - analogous to the way the dimension 3 case is proven.

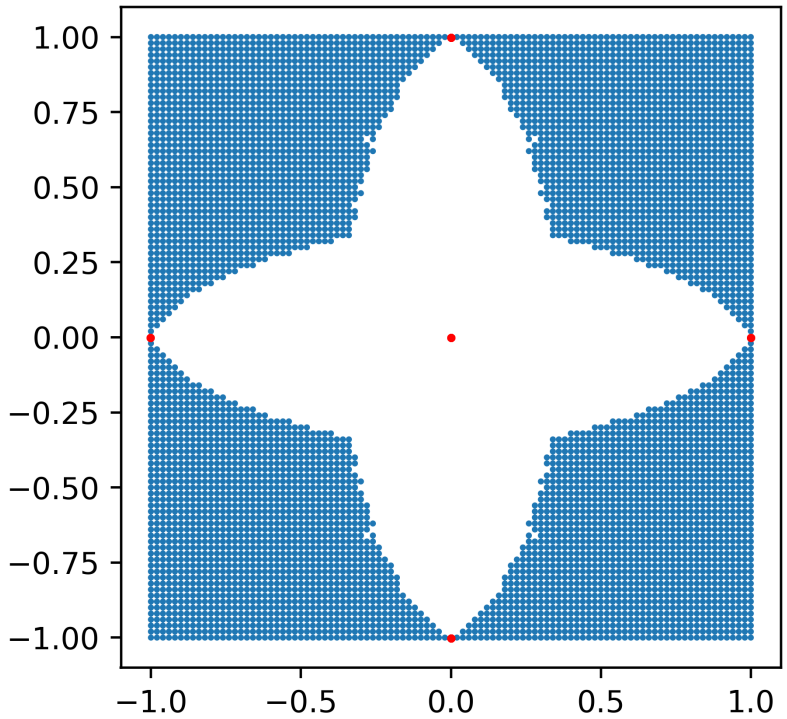


Figure: Point (x, y) is marked blue if $(x, y, 1)$ can be completed to a symmetric octahedron TC^* such that $\frac{1}{2}C_3 \subseteq TC_3^* \subseteq C_3$.

Moving on to $\mathbb{R}^4$ - pretty pictures

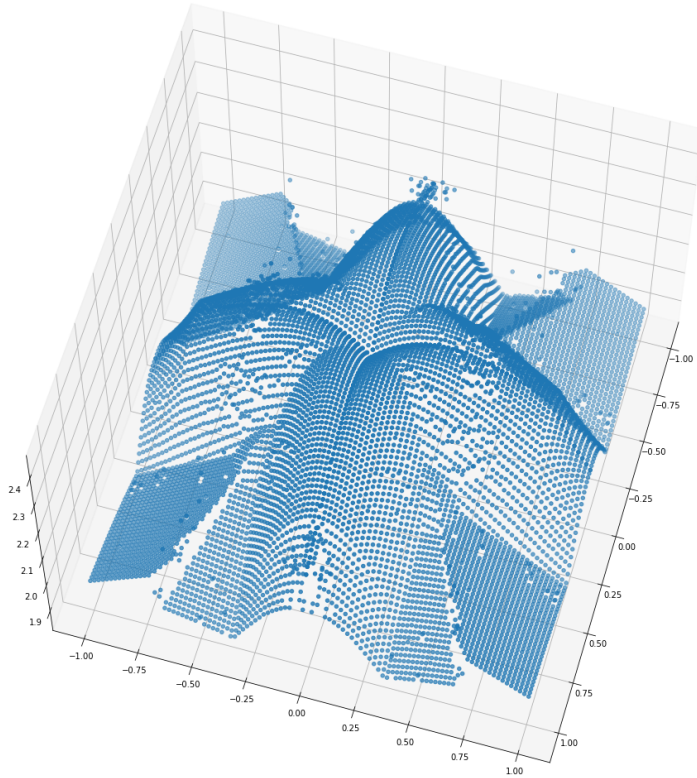


Figure: The z-axis is the lowest possible value $f(T)$ where $(x, y, 1)$ is a column of T.

The proof in dimension 4

We wish to prove that $d_{BM}(C_4, C_4^*) = 2$. The examples of operators T we already listed, give us the upper bound $d_{BM}(C_4, C_4^*) \leq 2$. With the aim of obtaining a contradiction we assume $\boxed{d_{BM}(C_4, C_4^*) < 2}$.

By assumption, there exists an operator T such that $rC_4 \subseteq TC_4^* \subseteq C_4$ where $r > \frac{1}{2}$.

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The proof in dimension 4

Definition 3 (String)

A 4-tuple $s = (c_1, c_2, c_3, c_4) \in \mathbb{R}^4$ is called a *string* if $|c_1| + |c_2| + |c_3| + |c_4| = 1$. Equivalently, *strings* are exactly the elements of ∂C_4^*. We will refer to both strings $s, -s$ as a *string pair*.

We will denote the rows of T as x, y, z, w so that $T = \begin{bmatrix} — & x & — \\ — & y & — \\ — & z & — \\ — & w & — \end{bmatrix}$.

Definition 4

For a string s and a row $b \in \{x, y, z, w\}$, we write $s \sim b$ if $|\langle s, b\rangle| \geq r$.

If s is a string, then the point Ts is on the boundary of TC_4^* which implies that $\|Ts\|_\infty \geq r$ or equivalently, $\exists b \in \{x, y, z, w\} : |\langle s, b\rangle| \geq r$. Therefore each string is associated with at least one row.

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The proof in dimension 4

Let $J := \{(\pm\frac{1}{4}, \pm\frac{1}{4}, \pm\frac{1}{4}, \pm\frac{1}{4})\}$ be the set of 16 strings.

Claim

Every row b is associated with at most 4 strings in J.

To see this, assume that $b = (b_1, b_2, b_3, b_4)$ and $b_1 \geq b_2 \geq b_3 \geq b_4 \geq 0$.
If b is associated with ≥ 5 strings, then it is associated with ≥ 3 string pairs. By comparison, among these pairs are certainly $\pm(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, $\pm(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4})$, and $\pm(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$.
Now we observe that

$$0 \leq \frac{1}{4}(b_1 + b_2 - b_3 + b_4) \leq \frac{1}{4}(b_1 + b_2) \leq \frac{1}{4} \cdot 2 = \frac{1}{2} < r$$

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Meaning that $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}) \nsim b$ which is a contradiction.

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The proof in diemnsion 4

To summarize, we have $J := \{(\pm\frac{1}{4}, \pm\frac{1}{4}, \pm\frac{1}{4}, \pm\frac{1}{4})\}$ as a set of 16 strings and 4 rows $\{x, y, z, w\}$. We know that each string in J is associated with at least one row, and we have just proven that each row is associated with at most 4 strings in J.

By counting, we conclude that:

- Each string in J is associated with exactly one row.
- Each row is associated with exactly 4 strings.

Again, for an example row $b = (b_1, b_2, b_3, b_4)$ where $b_1 \geq b_2 \geq b_3 \geq b_4 \geq 0$, these 4 strings are exactly $\pm(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ and $\pm(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4})$. Since $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}) \nsim b$, we conclude that $b_3 \neq b_4$ or $b_3 > b_4$. This allows us to make an important observation:

Observation

In each row $b \in \{x, y, z, w\}$, there exists an *unique* element with minimal absolute value in that row.

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The distribution of minimal elements in the matrix T

We now look at the entries of the matrix $T \in M_{4\times 4}(\mathbb{R})$. We will make the following identification:

- For each row, we will denote the unique element with the minimal absolute value as 0.
- For each row, the non minimal absolute value elements are nonzero, so we will denote the positive elements with $+$ and the negative elements with $-$.

Now it is important to note that we can perform several operations on the operator T satisfying $rC_4 \subseteq TC_4^* \subseteq C_4$ while preserving the inclusion. These include: Swapping rows, swapping columns, changing the sign of all elements in a column, changing the sign of all elements in a row.

The inclusion is also preserved when a column or a row is multiplied by a real number $s > 1$ assuming that after the multiplication each entry of T still satisfies $|T_{ij}| \leq 1$. So we can safely assume that each column contains an element with absolute value equal to 1.

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The distribution of minimal elements in the matrix T

We now wish to classify all the possible operators T satisfying $rC_4 \subseteq TC_4^* \subseteq C_4$ under the just defined identification and the mentioned symmetries.

We consider the number of zeroes (elements with minimal absolute value in their respective rows) in the column of T with most zeroes. Thus we have 4 cases:

- 4 zeroes in a column
- 3 zeroes in a column
- 2 zeroes in a column
- 1 zero in a column

In order to do the classification, we shall use the following two rules for any two different rows $a, b \in \{x, y, z, w\}$.

(i) The rows a, b do not have a matching layout in terms of signs.

(ii) If rows a, b have minimal elements in two different columns, then they do not match in the remaining two columns.

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For example of rule(i), we can't have two rows of the form $\begin{bmatrix} + & + & & \circ \\ + & + & \circ & \end{bmatrix}$, as they would be associated to the same string $\frac{1}{4}(1, 1, \mathrm{sgn}(a_3), \mathrm{sgn}(b_4))$ which is a contradiction.

Here we show how to apply the rules in the case of 3 zeroes in a column. In the end we rule out this case because rows y and z have a matching sign layout which is forbidden by rule(i)

$$\begin{bmatrix} + & + & + & \circ \\ + & & & \circ \\ + & & & \circ \\ + & & \circ & \end{bmatrix} \xrightarrow{\text{(ii)}[x,w]} \begin{bmatrix} + & + & + & \circ \\ + & & & \circ \\ + & & & \circ \\ + & - & \circ & \end{bmatrix} \xrightarrow[\text{(ii)}[z,w]]{\text{(ii)}[y,w]}$$

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By careful analysis, we obtain that either

$$T \approx \begin{bmatrix} + & + & + & 0 \\ + & - & 0 & + \\ + & 0 & - & - \\ 0 & + & - & + \end{bmatrix} \text{ or } T \approx \begin{bmatrix} + & + & + & 0 \\ + & + & - & 0 \\ + & - & 0 & + \\ + & - & 0 & - \end{bmatrix}.$$

We note that $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 0 & 1 & -1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & -1 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & -1 & 0 & -1 \end{bmatrix}$ are both examples of operators that achieve $\frac{1}{2}C_4 \subseteq TC_4^* \subseteq C_4$ which explains why we can't discard these cases as easily.
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Case 1

We assume $T \approx \begin{bmatrix} + & + & + & 0 \\ + & - & 0 & + \\ + & 0 & - & - \\ 0 & + & - & + \end{bmatrix}$.

We define the function $s_x : [0,1] \to \texttt{strings}$ by:

$$s_x(t) := \begin{cases} (\frac{1-t}{3}, \frac{1-t}{3}, \frac{1-t}{3}, -t) & x_4 \geq 0 \\ (\frac{1-t}{3}, \frac{1-t}{3}, \frac{1-t}{3}, t) & x_4 < 0 \end{cases}$$

Note that $\langle x, s_x(t) \rangle = \frac{1-t}{3}x_1 + \frac{1-t}{3}x_2 + \frac{1-t}{3}x_3 - t|x_4|$. We have

- $\langle x, s_x(\frac{1}{4}) \rangle = \frac{1}{4}x_1 + \frac{1}{4}x_2 + \frac{1}{4}x_3 - \frac{1}{4}|x_4| \geq r$ since $x \sim (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \pm\frac{1}{4}) \in J$
- $\langle x, s_x(\frac{1}{2}) \rangle = \frac{1}{6}x_1 + \frac{1}{6}x_2 + \frac{1}{6}x_3 - \frac{1}{2}|x_4| \leq 3 \cdot \frac{1}{6} = \frac{1}{2} < r$

$\implies \exists t_x \in [\frac{1}{4}, \frac{1}{2}) : \langle x, s_x(t_x) \rangle = r$. We see that $\langle x, s_x(t) \rangle \geq r$ for $t \leq t_x$.

Equivalently, $x \sim s_x(t)$ for $0 \leq t \leq t_x$.

We will call $\boxed{s_x(t_x)}$ the *specific string of row x*.

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Case 1

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The functions s_y, s_z, s_w and the numbers t_y, t_z, t_w are defined analogously to s_x and t_x. For example:

$$s_z(t) := \begin{cases} (\frac{1-t}{3}, -t, -\frac{1-t}{3}, -\frac{1-t}{3}) & z_2 \geq 0 \\ (\frac{1-t}{3}, t, -\frac{1-t}{3}, -\frac{1-t}{3}) & z_2 < 0 \end{cases}$$

and $t_z \in [\frac{1}{4}, \frac{1}{2})$ is a real number such that $\langle z, s_z(t_z) \rangle = r$. We have $z \sim s_z(t)$ for $t < t_z$ and $s_z(t_z)$ is the specific string of row z.

Case 1

Claim

For each $b \in \{x, y, z, w\}$, there is an another row $a \neq b$ such that $a \sim s_b(t_b)$. In other words, for all rows b, $s_b(t_b)$ is associated to some other row other than b.

We fix b. From the definitions of s_b and t_b, for small $\epsilon > 0$ we have $b \nsim s_b(t_b + \epsilon)$ (Since $\langle b, s_b(t_b + \epsilon)\rangle < r$).
Let $(\epsilon_n)_{n=1}^{\infty}$ be a sequence such that $\epsilon_n \to 0^+$. Since $b \nsim s_b(t_b + \epsilon_n)$, we have $a_n \sim s_b(t_b + \epsilon_n)\ \forall n$ where $a_n \neq b$. There are only 3 possible rows for a_n so one of the rows repeats infinitely often. Let this row be a. Now, there is a subsequence $(\epsilon_{n_k})_{k=1}^{\infty}$ such that $a \sim s_b(t_b + \epsilon_{n_k})\ \forall k$. This is equivalent to $|\langle a, s_b(t_b + \epsilon_{n_k})\rangle| \geq r$ so taking the limit as $k \to \infty$, we obtain $|\langle a, s_b(t_b\rangle)| \geq r$ or $a \sim s_b(t_b)$. □

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Case 1

We assume WLOG that $t_x = \min\{t_x, t_y, t_z, t_w\}$.
Now, $a \sim s_x(t_x)$ for some $a \neq x$. WLOG we take $a = y$ or $\boxed{y \sim s_x(t_x)}$ which gives us

$$\left|\frac{1-t_x}{3}y_1 + \frac{1-t_x}{3}y_2 + \frac{1-t_x}{3}y_3 \pm t_x y_4\right| \geq r \implies$$
$$\frac{1-t_x}{3}|y_1 + y_2| + \frac{1-t_x}{3}|y_3| + t_x y_4 \geq r \tag{2}$$

Since $t_x \leq t_y$, it follows that $\boxed{y \sim s_y(t_x)}$ or

$$\frac{1-t_x}{3}y_1 - \frac{1-t_x}{3}y_2 - t_x|y_3| + \frac{1-t_x}{3}y_4 \geq r. \tag{3}$$

Note that $y_1 - y_2 + |y_1 + y_2| \leq 2$. We now sum equations (2) and (3).

Case 1

We assume WLOG that $t_x = \min\{t_x, t_y, t_z, t_w\}$.
Now, $a \sim s_x(t_x)$ for some $a \neq x$. WLOG we take $a = y$ or $\boxed{y \sim s_x(t_x)}$ which gives us

$$\left|\frac{1-t_x}{3}y_1 + \frac{1-t_x}{3}y_2 + \frac{1-t_x}{3}y_3 \pm t_x y_4\right| \geq r \implies$$
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Case 1

By summing

$$\frac{1-t_x}{3}|y_1+y_2|+\frac{1-t_x}{3}|y_3|+t_x y_4 \geq r \text{ and}$$

$$\frac{1-t_x}{3}y_1-\frac{1-t_x}{3}y_2-t_x|y_3|+\frac{1-t_x}{3}y_4 \geq r$$

we obtain

$$\frac{1-t_x}{3}(y_1-y_2+|y_1+y_2|)-(t_x-\frac{1-t_x}{3})|y_3|+(\frac{1-t_x}{3}+t_x)y_4 \geq 2r.$$

Note that $t_x-\frac{1-t_x}{3}=\frac{4t_x-1}{3}\geq 0$ since $t_x \geq \frac{1}{4}$. Using $y_1-y_2+|y_1+y_2| \leq 2$ and $y_4 \leq 1$, we get

$$1=\frac{2(1-t_x)}{3}+(\frac{1-t_x}{3}+t_x) \geq LHS \geq 2r > 1.$$

We arrive at a contradiction and this finishes Case 1.

Case 2

In Case 2, $T \approx \begin{bmatrix} + & + & + & 0 \\ + & + & - & 0 \\ + & - & 0 & + \\ + & - & 0 & - \end{bmatrix}$.

We define the function $s_x : [0, \frac{1}{2}] \to \texttt{strings}$ by:

$$s_x(t) := \begin{cases} (\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - t, -t) & x_4 \geq 0 \\ (\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - t, t) & x_4 < 0 \end{cases}$$

Now $\langle x, s_x(\frac{1}{4}) \rangle \geq r$ since $x \sim (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \pm\frac{1}{4}) \in J$ and also

$$\langle x, s_x\left(\frac{1}{2}\right) \rangle = \frac{1}{4}x_1 + \frac{1}{4}x_2 - \frac{1}{2}|x_4| \leq \frac{1}{2} < r.$$

Therefore $\exists t_x \in [\frac{1}{4}, \frac{1}{2}) : \langle x, s_x(t_x) \rangle = r$. We will call $\boxed{s_x(t_x)}$ the *specific string of row x*.

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Case 2

So in Case 2, we define the specific string a bit differently but proceed in essentially the same way. Just like in Case 1, the specific string of row b is associated with some other row $a \neq b$.

We again WLOG assume that $t_x = \min\{t_x, t_y, t_z, t_w\}$ and have that $a \sim s_x(t_x)$ where $a \neq x$.

We need to consider possibilities:

- The row a has its zero in the same column as the row x. (or $a = y$)
- The row a has its zero in a different column as the row x. ($a \in \{z, w\}$)

By looking at the inequalities we get from $a \sim s_x(t_x)$ and $a \sim s_a(t_x)$ for each of these possibilities, Case 2 is finished by obtaining a contradiction in an analogous way to Case 1.

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Summary

To summarize, we have shown that if $rC_4 \subseteq TC_4^* \subseteq C_4$ for $r > \frac{1}{2}$, then we get a contradiction. This, along with our examples of optimal operators that achieve $r = \frac{1}{2}$:

$$T = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & -1 & -1 \\ 0 & -1 & 1 & -1 \\ 0 & 1 & 1 & -1 \end{bmatrix}, \begin{bmatrix} -1 & 1 & 0 & -1 \\ -1 & -1 & -1 & 0 \\ 1 & 0 & -1 & -1 \\ 0 & 1 & -1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \end{bmatrix}$$

proves that $\boxed{d_{BM}(C_4, C_4^*) = 2}$. □

References


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